

Housing units, household choices, and equilibrium prices

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The question. Why does the model offer houses of size 2, 4, 6 rather than 0.2, 0.4, 0.6, or 20, 40, 60? The numerical scale is a choice of units. It should not affect consumption, saving, tenure, fertility, or physical price responses. Our equations admit that equivalence, provided quantities, prices, and utility coefficients are converted together. Changing the grid alone would change the economic model.

What one housing unit means. The empirical quantity anchor is the American Housing Survey count of rooms, rather than bedrooms. The model uses that room scale for housing quantities and assumes services proportional to quantity, with a separate owner service premium. A four-room house is $H = 4$ in room units. Define $h = H/10$ if one housing unit instead represents ten rooms; the same house is then $h = 0.4$. Its physical size has not changed. The empirical room target must also be divided by ten when expressed in these new units.

Consumption and affordability. Income and consumption remain in the same monetary units. If rent per room is r , rent per ten-room unit is $10r$. The household therefore pays exactly the same amount:

$$rH = (10r)(H/10).$$

The same identity applies to purchase value, down payments, collateral, resale proceeds, depreciation, and property taxes. All these costs use price times housing quantity, with unchanged financing and tax rates. Consequently the same consumption and saving choices remain feasible. As an illustration, not a calibration estimate:

	Rooms	Ten-room units
Same dwelling's quantity	4	0.4
Rent per housing unit	10	100
Total rent paid	40	40
Income available	100	100
Consumption after rent and a child expense of 10	50	50

Fertility and utility. Preserving the budget is necessary but not sufficient: the value of consumption and housing must retain its comparison with the value of children. For a fixed consumption weight α , write material utility as

$$u^{\text{mat}}(c, H) = \frac{[c^\alpha H^{1-\alpha}]^{1-\sigma}}{1-\sigma}.$$

Here c is consumption and σ is curvature of utility over the consumption–housing composite. One transparent conversion replaces H by $10h$ in this expression. Material utility then stays exactly the same, and child utility needs no alteration.

Owners use housing directly; renters use a closed-form allocation in which the same unit conversion enters through rent. An equivalent conversion is therefore to keep its functional form in h and multiply *all utility-valued coefficients* by the common factor $K = (0.1)^{(1-\alpha)(1-\sigma)}$. With $\alpha = 0.733$ and $\sigma = 2$, $K = 1.849$. This includes child benefits, the fixed birth cost, bequest strength, and taste-shock scales. Total values then satisfy $V' = KV$, while the ratios of value differences to taste-shock scales remain unchanged. Birth-attempt and tenure probabilities, and optimal consumption and saving, are consequently unchanged. Re-estimation is not needed for this conversion.

Why the derivative can change without changing behavior. A derivative is measured per numerical housing unit. An upgrade from four to six rooms is $4 \rightarrow 6$ in room units and $0.4 \rightarrow 0.6$ in ten-room units. The numerical derivative and the numerical increment change together. The finite utility comparison for the same physical upgrade remains unchanged. For example, holding family state and tenure fixed and ignoring a housing floor, the consumption sacrifice at material indifference for $4 \rightarrow 6$ is 13.730% under the baseline weight 0.733. It is also 13.730% for $0.4 \rightarrow 0.6$ after a consistent conversion.

Housing supply and price curvature. Supply is expressed in the same housing units as demand. Let Q denote total housing, H_0 its supply scale, r unit rent, $\bar{r}$ reference unit rent, and ξ supply elasticity:

$$Q = H_0(r/\bar{r})^\xi, \quad r = \bar{r}(Q/H_0)^{1/\xi}.$$

Moving to ten-room units divides Q and H_0 by ten and multiplies r and $\bar{r}$ by ten. The equation remains satisfied at exactly the same physical allocation. Rescaling H_0 is part of this conversion; it does not require making H_0 an endogenous equilibrium variable.

Inverse supply is convex when $\xi < 1$. Nevertheless its proportional price response is constant:

$$\frac{d \log r}{d \log Q} = \frac{1}{\xi}.$$

At the specified $\xi = 0.630$, a 5% quantity expansion along the supply curve requires an 8.052% rent increase, in either set of units. The numerical slope changes with the coordinates; the implied change in rent per actual room does not. This calculation concerns the supply curve, not the full equilibrium response to a preference or fertility shock, which also depends on demand and the population closure.

Benchmark population one: how to determine H_0 . Keep rooms per household and the earnings-based monetary unit unchanged. Let $\bar{H}_b$ be housing demand per household at benchmark rent r_b . With benchmark household population normalized to one, housing-market clearing requires:

$$\bar{H}_b = H_0(r_b/\bar{r})^\xi \iff \boxed{H_0 = \bar{H}_b(\bar{r}/r_b)^\xi}.$$

For positive demand and rent, this coefficient always exists and makes aggregate housing supply equal demand. This is the precise accounting guarantee: the supply scale is consistent with the benchmark's price and dwelling choices. Birth replacement and fiscal balance remain separate equilibrium conditions.

If a saved solution instead reports population N^* , expressing that same solution with population one requires $H_0^{(1)} = H_0/N^*$ and dividing all aggregate population and housing stocks and flows by N^* . For the floor point on page 3, the full saved values give $H_0^{(1)} = 6.824$, rather than 6.294; supply becomes 5.977 rooms per normalized household, exactly its demand. Prices and household choices are unchanged by this population-unit conversion. An actual recalibration may change them. Neither operation guarantees a good empirical fit; the price and preferences must still be disciplined by data. Transitions must retain the resulting benchmark supply curve in the same units.

Application to the two housing preferences. The adopted reference's compensated family-dependent housing weights require its utility reference rent to convert with the grid. The floor experiment requires its physical floor to convert with the grid. These adjustments preserve the common utility factor K . The unit-conversion argument applies separately to both specifications; it does not prove that their calibration procedures are equivalent.

What the audit establishes. A read-only review and scalar arithmetic support the inspected budget, utility, choice and supply identities. They do not compare solved policies or explicitly evaluate the renter allocation. Source hashes cover the floor experiment, not the adopted reference. Full numerical equivalence remains unchecked, including feasibility cutoffs and regularizers. No model was imported, recalibrated, or solved. Services per room, elasticities and affordability require separate empirical validation.

From model prices to dollars: a worked example

The two units. Housing is measured in rooms; money is measured relative to the model’s baseline annual gross earnings scale, which is normalized to one. Let M be the number of 2007 dollars corresponding to one model monetary unit. Let p be the purchase price of one room and r its rent for one four-year model period. The model links them by $r = up$, where u is the period user-cost coefficient. Neither r nor u determines the earnings-to-dollar conversion.

Where the dollar scale comes from. The diagnostic uses PSID 2007 gross labor earnings of the reference person and spouse combined, not total household income. For reference persons aged 18–65 with nonnegative earnings and positive survey weights, it takes weighted means within twelve four-year age groups and then averages those twelve means equally, matching the model income profile’s age-cell normalization. The result is \$77,963.06 in 2022 purchasing power. Using the retained CPI-U annual indices converts this to 2007 dollars:

$$M \simeq 77,963.06 \times \frac{207.342}{292.655} \simeq \$55,235.75 \quad \text{per model monetary unit.}$$

This conversion is obtained from earnings, independently of the housing prices being checked; it was not chosen to make those prices match the AHS.

Purchase prices and rental payments. Purchase value is a stock, so multiply p by M . Rent is a payment over four years, so multiply r by M and divide by 48 months. Using the full saved reference values gives:

$$\begin{aligned} \text{Purchase value per room} &= 0.7898695 \dots \times M = \$43,629.03, \\ \text{Monthly rent per room} &= 0.1423972 \dots \times M/48 = \$163.86. \end{aligned}$$

The same conversion applied to the floor experiment gives the last column:

Cost per room	AHS 2007	Sept. 28 reference	Oct. 1 floor
Monthly rent (\$)	168.22	163.86	149.20
Purchase value (\$)	43,495.49	43,629.03	39,723.80

The data column uses weighted total annual contract rent divided by weighted renter rooms and by twelve; owner value uses weighted total reported dwelling value divided by weighted owner rooms. These are stock ratios, not averages of individual households’ price-per-room ratios.

What the reference rent $\bar{r}$ does. The supply equation uses a separate, fixed number $\bar{r} = 0.16$: $S(r) = H_0(r/\bar{r})^\xi$. Thus H_0 is the quantity supplied *at that reference rent*, not necessarily at the equilibrium rent r . On the same dollar conversion, $\bar{r}$ corresponds to \$184.12 per room per month; households need not pay that amount. It is not a price target or the earnings numeraire. It can be absorbed into the coefficient:

$$S(r) = Ar^\xi, \quad A = H_0/\bar{r}^\xi \simeq 19.966 \quad (H_0 = 6.294, \xi = 0.630).$$

This is the same supply curve. A third reference, r_* (code: `utility_reference_rent`), compensates family-dependent utility weights in the adopted reference; it is inactive in the floor specification. It is distinct from both actual rent r and supply reference rent $\bar{r}$.

Scope of the comparison. PSID earnings mainly concern the preceding tax year; AHS rents and self-reported values are not quality-adjusted. The floor point (no unsecured borrowing, chain 7, case 0173) remains experimental. Its preferences, entry/credit rules and calibration procedure differ; the price gap does not isolate one change. This preliminary check does not establish equivalence of the calibration procedures. The price-level readout and saved calculation retain full inputs and qualifications.